

Agentic Multimodal Models for Environmental Hyperspectral Unmixing

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Abstract. Hyperspectral unmixing is a key task in remote sensing that aims to decompose mixed pixels in hyperspectral images into their constituent material signatures, or endmembers, and their fractional abundances. We introduce an agentic Large Vision-Language Model (LVLM) as a refinement layer for a standard HySime–VCA–FCLSU unmixing pipeline. The agent combines spectral-library retrieval with visual inspection of abundance maps and iteratively refines the active endmember set through merge and discard operations, with constrained abundance estimation recomputed after each modification. Experiments on Jasper Ridge and HYDICE Urban show that the proposed approach substantially reduces both endmember-count error and matched reconstruction RMSE compared with the initial decomposition. These results indicate that LVLMs can effectively complement physically grounded hyperspectral unmixing methods as reasoning-driven refinement modules, without requiring direct end-to-end adaptation to hyperspectral data.

Keywords: Hyperspectral Unmixing · Multimodal Remote Sensing · Agentic Vision-Language Models

1 Introduction

Hyperspectral imaging records dense reflectance measurements over hundreds of narrow bands, enabling fine-grained and precise material discrimination [4]. Because limited spatial resolution and heterogeneous surfaces cause pixels to contain several materials, hyperspectral unmixing (HU) estimates the material spectral signatures, called *endmembers*, and their pixel-wise fractional abundances [3]. This sub-pixel representation supports environmental analysis, enabling the estimation of the abundance of elements such as vegetation, soil, water, and urban materials [7, 12].

Most HU methods use the linear mixing model because it is interpretable and compatible with abundance non-negativity and sum-to-one constraints [2, 10]. A critical challenge in this formulation is estimating the number of endmembers, which fixes the dimensionality of the problem. While standard benchmark

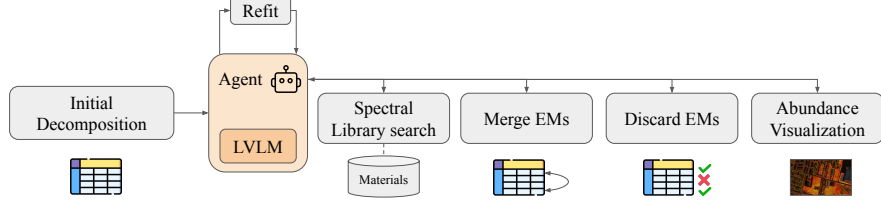


Fig. 1: Overview of the agentic refinement pipeline. The LVLM integrates spectral-library matches and abundance-map evidence to iteratively refine the active endmember set through merge and discard operations. After each modification, abundances are re-estimated using FCLSU.

evaluations often assume this number is known *a priori*, operational scenarios require estimating it from the data using signal-subspace methods, such as HySime [1]. Once the dimensionality is set, geometrical extraction methods like Vertex Component Analysis (VCA) [11] are standard baselines for identifying endmembers. Subsequently, abundance fractions are typically estimated via constrained optimization algorithms, such as Fully Constrained Least Squares Unmixing (FCLSU) [5]. However, these sequential pipelines remain sensitive to noise, highly mixed pixels, and the absence of pure signatures [13].

In this paper, we propose to employ a Large Vision-Language Model (LVLM) in an agentic configuration as a refinement layer in this challenging setting where the target-scene endmember count is unknown. LVLMs have recently been adapted to remote sensing for open-ended scene interpretation [6, 9], while agentic frameworks extend them with planning and external tools [15]. However, their application to HU remains limited due to the difficulty of applying LVLMs directly to the hyperspectral domain without additional fine-tuning. Building on a classical HySime-VCA-FCLSU decomposition, our approach can search a spectral library, visualize abundance maps, and modify the active endmember set via merge and discard operations. The key idea is to leverage the underlying LVLM’s physical knowledge and the visual backbone’s visual semantic understanding capabilities when fed with RGB data derived from the hyperspectral image. We evaluate the approach on Jasper Ridge [17, 18] and HYDICE Urban [14, 17], showing substantial improvements in estimated endmember count and reconstruction error.

2 Methodology

Problem formulation and initialization. Let $\mathbf{Y} \in \mathbb{R}^{L \times N}$ be the hyperspectral image, containing N pixels, each carrying L bands. Under the linear mixing model,

$$\mathbf{Y} = \mathbf{M}\mathbf{A} + \mathbf{N}, \quad \mathbf{A} \geq 0, \quad \mathbf{1}^\top \mathbf{A} = \mathbf{1}^\top, \quad (1)$$

where $\mathbf{M} \in \mathbb{R}^{L \times K}$ contains the endmembers and $\mathbf{A} \in \mathbb{R}^{K \times N}$ their abundances. Because K is unknown, HySime [1] estimates an initial dimension K_0 , VCA [11]

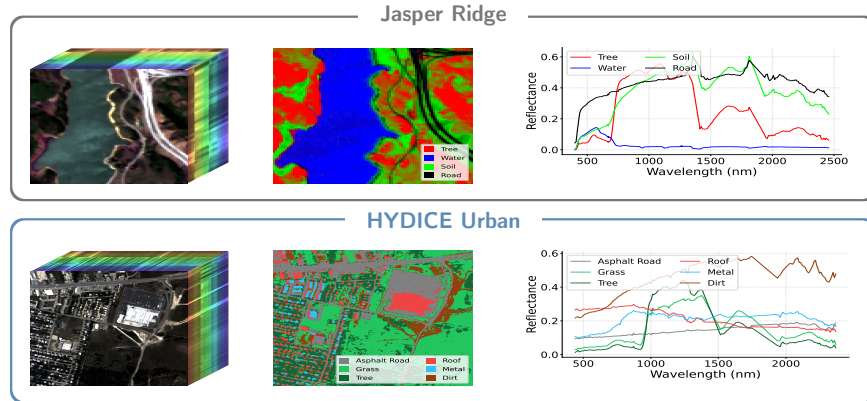


Fig. 2: Hyperspectral unmixing benchmarks. Jasper Ridge (top) and HYDICE Urban (bottom). Each row shows the false-color image, reference abundance maps, and corresponding spectral signatures.

extracts $\mathbf{M}_0$, and FCLSU [5] estimates $\mathbf{A}_0$. This deliberately simple initialization enables the proposed agent to serve as a plug-and-play refinement layer.

Agentic refinement. The LVLM controls a ReAct-style tool-using loop [16], summarized in Fig. 1. The agent’s working memory is represented by a compact table containing endmember identifiers, candidate material labels, cluster mass, and active/discarded status. Spectral vectors and abundance arrays are manipulated only by tools, avoiding long numerical prompts, since they are not directly interpretable as text by the underlying LVLM.

Specifically, the agent has access to four tools. (i) `library_search` compares an active spectrum against the USGS material library [8] using spectral angle distance, returning the most relevant material matches along with their textual categories. (ii) `compute_abundance` renders the current FCLSU abundance map as a heatmap blended with the scene RGB image, providing spatial evidence about coherence and redundancy to the LVLM. (iii) `merge_endmembers` replaces two spectra with their mass-weighted average, combines their candidate labels, and refits FCLSU. (iv) `discard_endmember` removes a weak or redundant component and likewise refits the abundance estimator.

At each turn, the agent selects an action based on the accumulated library matches, visual evidence, and the current state. The process stops when the active set is judged compact and stable by the agent, returning the final endmembers, abundance maps, and the decision log.

3 Experimental Setting

Datasets and implementation. We employed two different datasets (Fig. 2). Jasper Ridge [17, 18] uses a 100×100 region with 198 retained bands and four reference materials: road, soil, water, and trees. HYDICE Urban [14, 17] contains

307×307 pixels and 162 retained bands; we use the six-material annotation comprising asphalt, grass, trees, roofs, dirt, and metal. The agent uses Qwen-3.6 27B with reasoning disabled and temperature zero. Its retrieval tool accesses the USGS Spectral Library [8]. Abundance heatmaps are overlaid on the RGB rendering with 50% opacity. Because VCA is stochastic, results are reported as mean and population standard deviation over ten runs.

Metrics. We report endmember-count error $\Delta K = |K_{\text{est}} - K_{\text{gt}}|$ and *Matched RMSE* of the image reconstruction, obtained by optimally associating estimated endmembers with the reference ones. In the unknown- K setting, Matched RMSE is preferred because it only considers the abundance maps obtained from the subset of K_{est} endmembers most affine to the ground-truth materials.

3.1 Results and Discussion

Table 1 reports the quantitative results on HYDICE Urban and Jasper Ridge.

Table 1: Results on HYDICE Urban and Jasper Ridge. Values are mean $\pm$ population standard deviation over 10 independent runs.

Method	HYDICE Urban		Jasper Ridge	
	ΔK	Matched RMSE	ΔK	Matched RMSE
Initial	21.0 ± 0.0	0.157 ± 0.007	13.0 ± 0.0	0.121 ± 0.009
LVL Agent	6.0 ± 6.1	0.079 ± 0.052	1.2 ± 0.7	0.027 ± 0.007

On HYDICE Urban, the initial decomposition strongly overestimates the number of materials, with $\Delta K = 21.0$. The LVL agent obtains a strong reduction on the same scene, decreasing ΔK to 6.0 ± 6.1 and improving Matched RMSE to 0.079 ± 0.052 . A similar trend is observed on Jasper Ridge. The initial decomposition yields $\Delta K = 13.0$ and a Matched RMSE of 0.121 ± 0.009 . The LVL agent reduces the endmember-count error to 1.2 ± 0.7 and substantially improves Matched RMSE to 0.027 ± 0.007 . Overall, the results show that the LVL agent, on both datasets, substantially reduces the endmember-count error and the matched reconstruction RMSE, indicating that the retained components are more closely aligned with the reference material representation.

4 Conclusion

In this work, we presented an LVL agent that refines hyperspectral unmixing through spectral-library retrieval, abundance-map inspection, and iterative merge or discard operations. On Jasper Ridge and HYDICE Urban, it substantially improves endmember-count estimation and matched reconstruction accuracy relative to the HySime, VCA, and FCLSU initialization. These results support foundation models as reasoning-driven refinement modules rather than end-to-end replacements for physically grounded HU algorithms.

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